

Library Book Management System - Document

1. Introduction

This project is a simple Library Book Management System developed in Python.

It maintains a list of books and allows the user to perform operations like adding, viewing, issuing, returning, searching, and deleting books.

2. Features

Add a new book

View all books

Issue a book

Return a book

Search book by title

Remove/Delete book

Exit the program

3. Data Structures Used

books = [] - Stores all book details in a list.

book_id_counter = 1 - Helps assign unique IDs to books.

4. Function Explanations

4.1 add_book()

Adds a new book into the list with an auto-incremented ID.

4.2 view_books()

Displays all books in a formatted structure.

4.3 issue_book()

Allows the user to issue a book if it's available.

4.4 return_book()

Marks a previously issued book as available.

4.5 search_book()

Searches for a book using a partial or full title.

4.6 remove_book()

Deletes a specific book by its ID.

5. Main Program Loop

Displays options and executes functions based on user input.

6. Conclusion

This project uses lists, dictionaries, functions, loops, and conditions to create a basic library management system.